LOCALLY NOETHERIAN DECENT SPACES ARE QUASI-SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Diagonal of locally quasi-finite morphisms. Morphisms of Spaces, lemma diagonal-unramified-morphism (cross-chapter, named). The diagonal is unramified.
- Quasi-separatedness over schemes. spaces-properties lemma quasi-separated-over-scheme (cross-chapter, named). It suffices to check the diagonal after an etale cover.

Source reference. Original reference 086R: focused-seeds/geometry/S9-decent-locally-noetherian-decent-quasi-separated/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07RY: focused-seeds/geometry/S9-decent-locally-noetherian-decent-quasi-separated/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AHR: focused-seeds/geometry/S9-decent-locally-noetherian-decent-quasi-separated/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Any locally Noetherian decent algebraic space is quasi-separated.*

Proof. Namely, let X be an algebraic space (over some base scheme, for example over $\mathbf{Z}$) which is decent and locally Noetherian. Let $U \to X$ and $V \to X$ be étale morphisms with U and V affine schemes. We have to show that $W = U \times_X V$ is quasi-compact (Properties of Spaces, Lemma 0AHR). Since X is locally Noetherian, the schemes U, V are Noetherian and W is locally Noetherian. Since X is decent,

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the fibres of the morphism $W \to U$ are finite. Namely, we can represent any $x \in |X|$ by a quasi-compact monomorphism $\mathrm{Spec}(k) \to X$. Then U_k and V_k are finite disjoint unions of spectra of finite separable extensions of k (Remark 03II) and we see that $W_k = U_k \times_{\mathrm{Spec}(k)} V_k$ is finite. Let n be the maximum degree of a fibre of $W \to U$ at a generic point of an irreducible component of U. Consider the stratification

$$U = U_0 \supset U_1 \supset U_2 \supset \ldots$$

associated to $W \to U$ in More on Morphisms, Lemma 086R. By our choice of n above we conclude that U_{n+1} is empty. Hence we see that the fibres of $W \to U$ are universally bounded. Then we can apply More on Morphisms, Lemma 07RY to find a stratification

$$\emptyset = Z_{-1} \subset Z_0 \subset Z_1 \subset Z_2 \subset \ldots \subset Z_n = U$$

by closed subsets such that with $S_r = Z_r \setminus Z_{r-1}$ the morphism $W \times_U S_r \to S_r$ is finite locally free. Since U is Noetherian, the schemes S_r are Noetherian, whence the schemes $W \times_U S_r$ are Noetherian, whence $W = \coprod W \times_U S_r$ is quasi-compact as desired. $\square$